

Series 3

$$1) \quad u_1 = 56+3 \quad a = 6t+3 \\ u_n = 176+11 \quad = 56+3+4n-4$$

$$12t+12 = 4n \\ n = 3t+3$$

$$S_n = \frac{n}{2} [2a + (n-1)d] \\ = \frac{3t+3}{2} [2(56+3) + 4(3t+2)] = \frac{3t+3}{2} (106+6+12t+8) \\ = (3t+3)(11t+7)$$

Let t be even.
Then $t = 2n \quad | \quad n \in \mathbb{R}$

$$\text{And } S_n = (6n+3)(22n+7) = 132n^2 + 108n + 21 \quad | \quad \text{not divisible by 12}$$

Then let t be odd
Then $t = 2n+1 \quad | \quad n \in \mathbb{R}$

$$\text{And } S_n = (6n+6)(22n+18)$$

$$= 132n^2 + 240n + 108 \\ = 12(11n^2 + 20n + 9)$$

$\therefore$ divisible by 12.

$$2) \quad a) \quad n_1 = 11 \log_2 1024 = 11$$

$$b) \quad r_2 = 4 \\ n_2 = 6$$

$$c) \quad \text{Let } k=2$$

$$\text{Then } r_3 = \sqrt{2} \\ n_3 = 21$$

$$\therefore S_n = \frac{1 - (r_3)^{n_3}}{1 - r_3}$$

$$= \frac{1 - (\sqrt{2})^{21}}{1 - \sqrt{2}} \\ = \frac{1 - 1024\sqrt{2}}{1 - \sqrt{2}}$$

$$\begin{aligned} &= \frac{-(1 - 1024\sqrt{2})(1 + \sqrt{2})}{-1} \\ &= -(1 - 1023\sqrt{2} - 2048) \\ &= 2047 + 1023\sqrt{2} \end{aligned}$$